Noé Jager

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SUMMARY

ML/AI Systems Engineer with strong experience in transformer architectures, representation learning, and end-to-end system design. Winner of HackOS4 (1st place), AI Hackathon organised by McGill University (Montréal, Canada),, and team lead for multi-task robotic learning research. Skilled in PyTorch, HuggingFace, LangChain, Neo4j, and modular ML engineering.

SKILLS

- **Languages:** Python, SQL, TypeScript
- **Frameworks:** PyTorch, HuggingFace, LangChain, Neo4j, LeRobot
- **ML Concepts:** Transformers, Representation Learning, RAG, LLM Agents
- **Engineering:** Data Pipelines, Experiment Tracking, OOP for ML
- **Tools:** MLflow, Optuna, Git, AWS (EC2, Cloudfront, Amplify), Docker

PROJECTS & EXPERIENCE

Research Internship at AppOX — May 2025 - now

Developed a benchmarking framework to evaluate privacy risks in synthetic tabular datasets. Compared baseline against representation-learning based attacks (contrastive learning, VAE), demonstrating faster detection performance at scale.

Tools & methods: Python, PyTorch, MLflow, Optuna, VAE, Contrastive Learning, Data Privacy Metrics

Behavior-1k Challenge: Multi-Task Robotic Learning in Simulation — December 2025 – now

Topic: Fine-tuning vision-language action (VLA) models to build a generalist robotics agent for multi-task simulated household tasks.

- Led the team in fine-tuning open-source **VLA models** on the **Behavior-1k dataset** using the **LeRobot** platform.
- Built a multi-task generalist agent capable of solving diverse simulated household tasks.
- Evaluated performance using simulated task success metrics and benchmark comparisons.
- Collaborated with the RWTH Robotics Club to ensure scalable robot learning pipelines.

Tools & methods: Python, Hugging Face, LeRobot, VLA models

Neural-Blueprints - Modular Deep Learning Framework — November 2025 - now

Topic: A Python package designed for rapid prototyping of deep learning models with clean, modular architecture components and visualization tools.

- Built reusable architectures: **MLPs, Bert-Transformers, Auto-encoders, and more**.
- Designed **composable components** (attention, conv, dense, tabular/linear/multimodal projections).
- Created a **typed configuration system** with validation for architectures and training pipelines.
- Implemented **tabular preprocessing tools** and **tabular dataset variants** for various tasks.
- Developed **training utilities**: Losses, metrics, and visualization helpers.

Tools & methods: Python, PyTorch, OOP design, modular ML systems, tabular ML

McGill HackOS4-2025: Single-Cell Perturbation Challenge – 1st Place — March – April 2025

Topic: A ML/biotech hackathon from McGill University aiming at predicting changes in Proteins RNA from drugs

- Predicted RNA perturbations using a custom **Transformer Encoder + 1D CNN** architecture.
- Integrated **ChemBERTa embeddings** and cell-type encodings.
- Applied **Dimensionality reduction** via Truncated SVD on high-dimensional gene data.
- Performed **Molecular augmentation via isomers** to generalize to unseen molecules.

Tools & methods: Python, PyTorch, HuggingFace, Transformer Encoder

GraphRAG Chatbot application using Neo4j, Langchain, React and Flask — January - March 2025

Topic: A full-stack LLM assistant leveraging knowledge graphs to store and retrieve long-term user memories.

- Applied Generative AI and **GraphRAG** concepts to create **multiple agents** for response generation and database updates.
- Built a Flask server to execute agents with low latency (around **5–6 seconds** response time).
- Designed and prototyped the client UI in **Figma** and developed the frontend using **React + Redux**.
- Implemented automated Neo4j backup and a request-locking mechanism for reliability and consistency.

Tools & methods: Python, Flask, Neo4j, Langchain, React, Redux, LLM agents, RAG and knowledge graphs

EDUCATION

Master in Computer Science at University of Montréal — September 2022 -Now

Bachelor of Computer Science at the University of Luxembourg — September 2019 - July 2022